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The Impact of Time-Variant Migration Aperture TVMA on Seismic Image Quality Enhancement: A Case Study from a Complex Structural Oil Field in the Eastern Desert

M. S. Abdelshakor, Specialist Geophysicist, UEE, Cairo, Egypt; Tarek Talkhan, Geophysics Lead, UEE, Cairo, Egypt; M. Metwally, Time Processing Manager, PTG, Cairo, Egypt

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Abstract

The Area A oil fields, located in the Eastern Desert, represent the western onshore extension of the Gulf of Suez (GOS) basin. The geology of these fields is characterized by significant subsurface complexity, including complex fault networks, highly variable dips, and strong velocity heterogeneity. Given these challenges, this study focuses on enhancing seismic imaging resolution and accuracy in Area A. Conventional seismic migration techniques often use fixed apertures, which struggle to adapt to the highly variable dip, depth, and velocity conditions observed vertically across the area. This limitation leads to migration artifacts, poor focusing, and reduced signal-to-noise (S/N) ratios. To address this, we developed and applied a time-variant migration aperture (TVMA) optimization strategy tailored to these complex geological conditions. A comparative assessment against conventional fixed-aperture migration methods was conducted to evaluate improvements in structural definition.

The workflow began with the derivation of smoothed stacking velocities from a detailed 500 m × 500 m velocity analysis grid, which served as the initial velocity model for migration. Kirchhoff pre-stack time migration (PSTM) was selected due to its effectiveness in handling smooth velocity variations in both spatial and temporal domains. The core innovation of this study was the integration of a TVMA approach within the Kirchhoff PSTM framework. Optimal aperture functions were determined through iterative testing, and analysis, with the goal of maximizing constructive interference of reflection energy while minimizing migration artifacts and noise. The optimized TVMA PSTM results were systematically compared with those from a conventional fixed-aperture PSTM using identical input data and velocity models.

The optimized TVMA PSTM technique significantly improved seismic imaging compared to the fixed-aperture approach. Key enhancements included:

- Improved focusing of reflection energy, particularly along steeply dipping faults and complex structures.
- Reduced migration swing artifacts and background noise, leading to a higher S/N ratio.

- Clearer subsurface structural definition, with better reflector continuity and improved delineation of subtle structural features.

The novelty of this work lies in the specific adaptation and optimization of a time-variant migration aperture strategy to address the unique imaging challenges posed by the complex geology of the Eastern Desert oil field. While TVMA concepts exist, this study demonstrates a practical workflow for deriving and applying optimized aperture functions directly informed by the characteristics of the local dataset and geological environment.

Introduction

Seismic imaging stands as a cornerstone of modern hydrocarbon exploration and development, providing the fundamental means for interpreting subsurface geological structures. The primary objective of seismic migration, a critical step in seismic data processing, is to accurately re-locate seismic events from their recorded positions on the surface to their true subsurface spatial coordinates. This process is essential for achieving increased spatial resolution and effectively resolving complex geological features, thereby transforming raw seismic echoes into a geologically meaningful subsurface image. Migration fundamentally converts information initially recorded as a function of time into precise spatial features, both laterally and vertically, allowing for a more accurate representation of the subsurface.

The Migration Method used in this study is Kirchhoff Pre-Stack Time Migration (PSTM), which is a widely utilized and robust method in seismic data processing, particularly effective for handling smoothly varying velocity fields in both spatial and temporal domains. A critical parameter in the implementation of Kirchhoff migration, and indeed any migration technique, is the migration aperture. This term refers to the spatial extent of the input data that is summed to form a single output migrated trace, and its proper selection is crucial for obtaining high-resolution images of subsurface geological structures.

Many hydrocarbon-bearing regions worldwide, including the Area A oil fields in the Eastern Desert, are characterized by significant subsurface complexity. This complexity manifests as intricate fault networks, severely variable dips, and pronounced velocity heterogeneity within the geological formations. These geological conditions present formidable challenges for seismic imaging. Conventional seismic migration techniques, which frequently employ fixed apertures, struggle to adapt to the highly dynamic dip, depth, and velocity conditions that vary significantly both vertically and laterally across such areas.

The inherent inability of a fixed aperture to adapt to these rapidly changing subsurface properties directly leads to several detrimental effects on seismic image quality. For instance, the severely variable dips and strong velocity heterogeneity mean that the optimal spatial extent for summing seismic energy changes dramatically with depth and across different structural elements. A fixed aperture, by its very nature, cannot accommodate these dynamic conditions. This fundamental mismatch between a static processing parameter and dynamic subsurface properties is the direct cause of observed migration artifacts, poor focusing of seismic energy, reduced signal-to-noise (S/N) ratios, and smeared events. The presence of thick evaporite sequences, for example, further exacerbates these issues by causing significant attenuation of seismic energy and introducing dip reversals, which conventional fixed-aperture methods cannot adequately address. This establishes a clear, direct causal link between the complex geological environment and the limitations encountered with traditional seismic imaging approaches.

To overcome the inherent limitations of fixed-aperture methods in complex geological settings, this study introduces and applies a Time-Variant Migration Aperture (TVMA) optimization strategy. This innovative approach is specifically tailored to the complex geological conditions prevalent in Area A, aiming to dynamically adjust the migration aperture throughout the seismic section.

Geological Context and Imaging Challenges of Area A

The Area A oil fields constitute a strategic onshore asset in Egypt's Eastern Desert situated within the unstable shelf of the North African Craton and representing the western extension of the Gulf of Suez (GOS) rift basin ([Figure 1](#)).



Figure 1—Satellite Base Map shows the location of Area A fields in the western onshore extension of the Gulf of Suez.

The Gulf of Suez Rift itself is a significant continental rift zone, which experienced active rifting from the Late Oligocene (approximately 28 million years ago) to the end of the Miocene (around 5 million years ago). This basin is a crucial strategic transit hub for oil reservoirs, connecting the Red Sea and the Mediterranean Sea, highlighting the economic importance of accurate subsurface imaging in the region.

The fundamental geological framework of the region is characterized by a Precambrian basement, which is part of the Arabian-Nubian Shield. This basement comprises complex crystalline igneous and metamorphic rocks, including gneisses, volcanics, metasediments, and various intrusive bodies like granites, granodiorites, and dolerite dykes. Overlying this basement are extensive Paleozoic and Mesozoic sedimentary sequences. These include Cambrian sandstones, Lower Carboniferous dolomitic and shaly units, and sandstones.

The Mesozoic succession features the Triassic Formation and the Upper Jurassic to Lower, often informally referred to as the 'NubianU' sandstone. Furthermore, the Miocene sequences are particularly vital, as they provide a crucial record for understanding the syn-rift structural and stratigraphic evolution of this continental rifting system. [[Figure 2](#)].

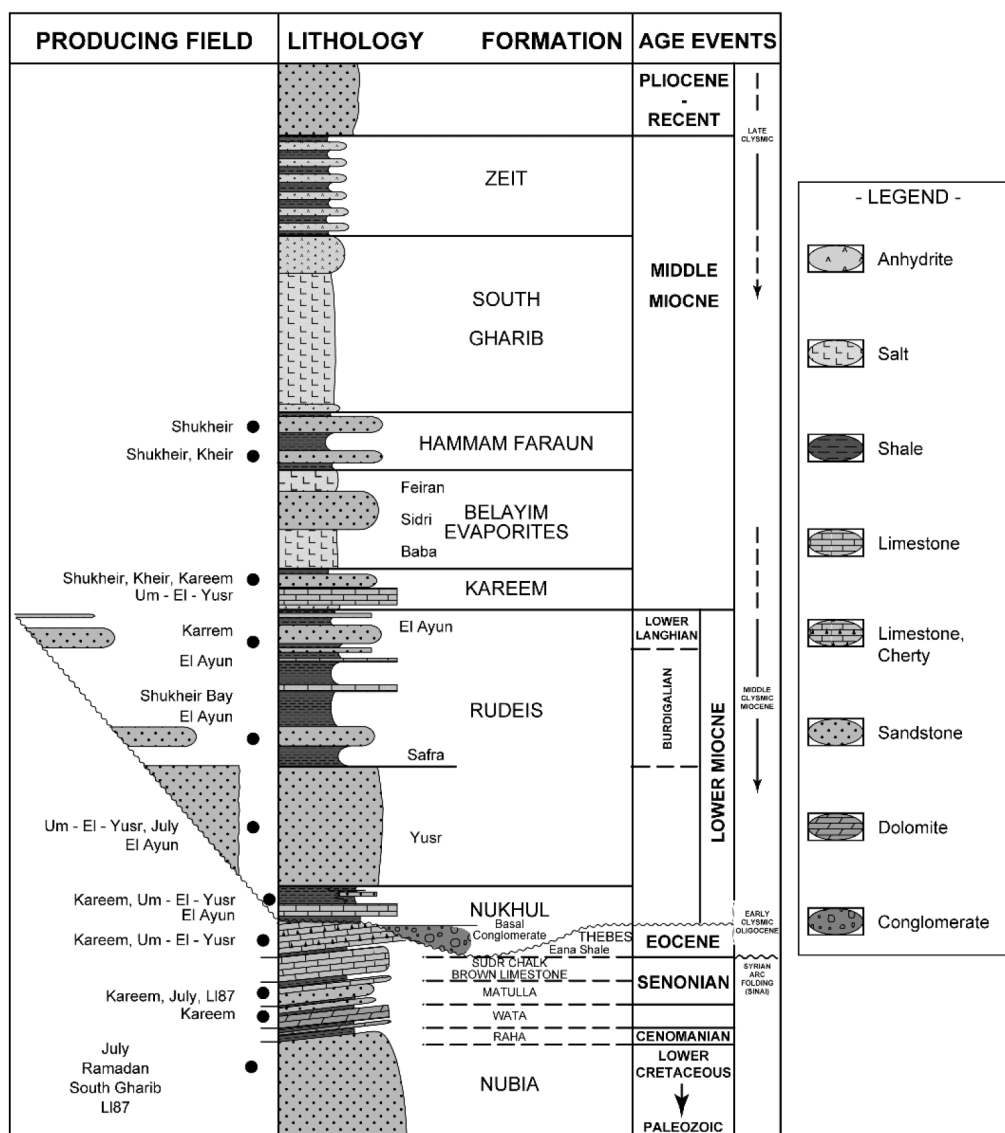


Figure 2—A summarized stratigraphic column of Area A fields lithologic succession

The Gulf of Suez is renowned as a sedimentary rift-basin characterized by highly complex structural settings. The extensional plate motion during the Miocene epoch led to the formation of four enormous half-graben structures, each featuring intricate faulting patterns. These structures are associated with significant fault networks that exhibit severely variable dips, including dip reversals in multiple orientations, which significantly affect seismic data interpretation. Hydrocarbon accumulations within these fields are notably associated with horst fault blocks, underscoring the importance of accurately delineating these complex structural traps. The challenges observed in Area A are not isolated; they are representative of the broader Gulf of Suez basin. This regional significance implies that the solutions developed for Area A could have wider applicability and impact in similar complex rift basins globally, extending the relevance of this research to a larger industry audience [Figure 3]

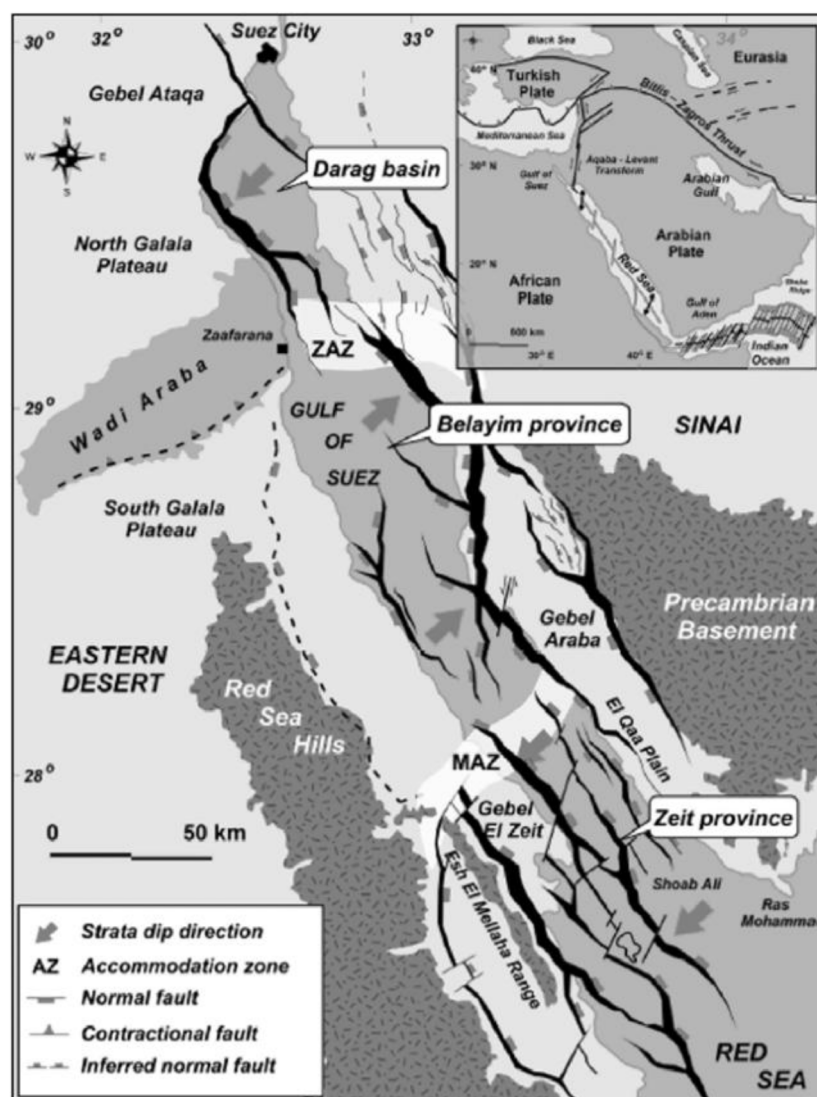


Figure 3—Tectonic map of the Gulf of Suez region representing major fault trends and accommodation zones adapted from (Bosworth and McClay, 2001; El Diasty et al., 2020; Radwan et al., 2020b).

A major impediment to achieving high-quality seismic images in the Gulf of Suez area is the pervasive presence of thick near-surface evaporites, particularly the Miocene Evaporite Sections. These evaporite layers are a direct and specific cause of several critical imaging problems. They induce significant attenuation and weakening of seismic energy as waves propagate through them, which in turn masks faint reflections originating from deeper, pre-Miocene interfaces. This phenomenon leads to inadequate transmission of seismic energy, often resulting in limited or no energy reaching the deeper strata of interest.

Beyond evaporites, the presence of thick Rudeis Clastic also contributes to the absorption and scattering of seismic energy, further degrading data quality. Moreover, enclosed flaws within the subsurface can lead to additional diffraction, scattering of noise, and the generation of unpredictable "fake reflections," complicating the seismic image. These factors contribute to a highly heterogeneous velocity field that poses a substantial challenge for conventional seismic processing.

The confluence of complex faulting, highly variable dips, and pronounced velocity heterogeneity—driven by thick evaporites and clastics—creates a challenging environment for seismic imaging in Area A.

The following table summarizes these key geological challenges and their direct impact on seismic imaging.

Table 1—Key Geological Challenges and Their Seismic Manifestations in Area A

Geological Feature	Seismic Manifestation	Impact on Conventional Fixed-Aperture Migration
Complex Fault Networks	Diffraction tails, Poor reflector continuity	Migration artifacts, Difficulty in structural delineation
Severely Variable Dips	Dip reversals, Poor reflector continuity	Poor focusing, Reduced S/N ratio, Migration artifacts
Thick Miocene Evaporites	Attenuation/Weakening of seismic energy, Masking of faint reflections	Limited energy penetration, Inaccurate interpretation
Rudeis Clastics	Absorption/Scattering of energy	Reduced S/N ratio, Poor focusing

Kirchhoff Pre-Stack Time Migration and its Main Considerations

Kirchhoff migration is a widely adopted seismic imaging technique rooted in the diffraction summation approach, which is a practical and robust implementation of Huygens' principle. The fundamental concept involves treating each subsurface imaging point as a potential scatterpoint.

For each such point, a theoretical diffraction shape is computed. The migration process then involves summing and weighting the amplitudes of the input seismic energy along this calculated diffraction path. The resulting summed energy is subsequently placed at the corresponding scatterpoint location on the migrated seismic section. This method is considered a high-frequency approximation and is particularly effective for handling smoothly varying velocity fields and imaging steeply dipping geological structures.

The theoretical foundation of Kirchhoff migration is rigorously derived from the integral solution to the scalar wave equation. In its general discrete form, the Kirchhoff integral for migration can be expressed as:

$$P_{out} = C \sum_A W(s, r, m) \left(\frac{\partial}{\partial t} \right)^n U(s, r, t_D)$$

In this equation, P_{out} represents the migrated output image, C is a constant scaling factor, and A denotes the migration aperture, which is the spatial summation range. The term $W(s, r, m)$ is an amplitude weighting function that incorporates critical physical factors such as the obliquity factor (cosine of the angle of propagation) and the spherical spreading factor (inversely proportional to distance).

The operator $(\partial/\partial t)^n$ represents a trace filter, often referred to as a rho filter, which applies necessary phase and amplitude corrections to the seismic wavelet. $U(s, r, t_D)$ is the input seismic data, and t_D is the two-way diffraction traveltimes.

For three-dimensional (3D) Pre-Stack Time Migration (PSTM), the two-way diffraction traveltimes (t_D) is typically calculated using the double-square-root (DSR) formula:

$$t_D = \sqrt{t_0^2/4 + |s - m|^2/v^2} + \sqrt{t_0^2/4 + |r - m|^2/v^2}$$

Here, t_0 represents the two-way vertical traveltime at the imaging point, s and r denote the source and receiver locations, respectively, m is the projected surface location of the imaging point, and v is the root-mean-square (RMS) velocity at the imaging point. The RMS velocity, specifically at the output time sample (which corresponds to the apex time τ of the hyperbola), is crucial for accurately defining the hyperbolic summation path along which amplitudes are summed.

Main Considerations

Velocity Model. The accuracy of the velocity model used in Kirchhoff PSTM is paramount, as it directly governs the correct positioning of reflectors and the proper focusing of the final seismic image. PSTM typically operates with a Root-Mean-Square (RMS) velocity field, denoted as $VRMS(x,y,t)$, which is allowed to vary both vertically (with time/depth) and laterally (across the survey area).

While PSTM is known for its robustness to certain levels of velocity model error, its performance can degrade significantly in the presence of strong lateral velocity variations. Consequently, most migration algorithms, including Kirchhoff PSTM, necessitate the use of a smooth velocity function to ensure stable and accurate results. [Figure 4].

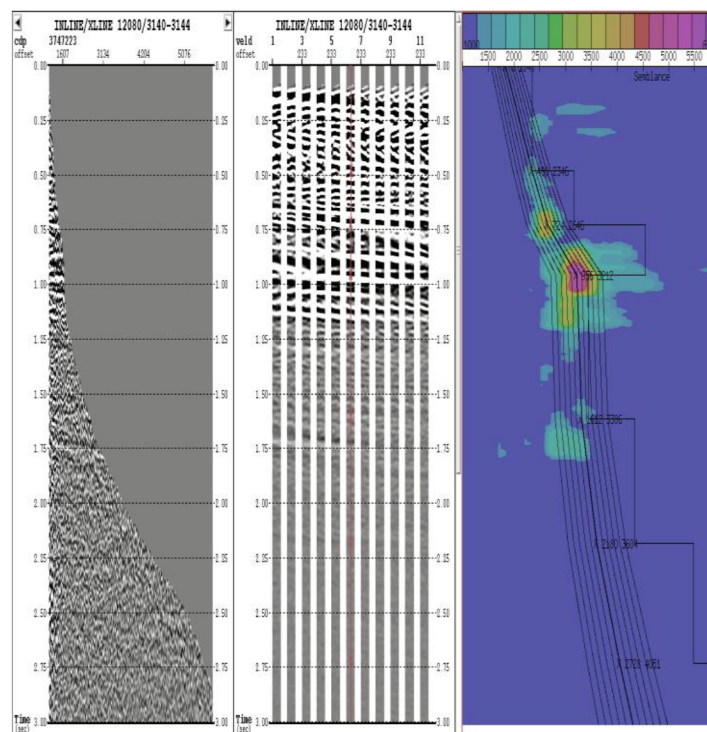


Figure 4—Velocity Analysis with CDP Gathers

Migration aperture. Migration aperture defines the spatial (horizontal) and temporal (vertical) extent of the input seismic data that contributes to the calculation of a single output migrated trace. This parameter is critical because it determines how much of the recorded seismic energy is included in the summation process for each subsurface point. It is essential for accurately capturing and correctly positioning dipping events and diffractions, which are fundamental to forming a high-resolution image of subsurface features.

The choice of aperture size has a profound impact on the final image quality:

- **Small Aperture:** An aperture that is too small can lead to the destruction of steeply dipping events, resulting in a poor overall image. It tends to pass only flat or nearly flat events, effectively filtering out valuable information from complex structures. Furthermore, excessively small apertures can

inadvertently organize random noise into spurious, horizontally dominant events, particularly in deeper sections, thereby degrading the signal-to-noise ratio.

- **Large Aperture:** Conversely, an excessively large aperture, while potentially capturing more energy and improving image quality, significantly increases computational cost and processing time unnecessarily. Moreover, a large aperture can degrade the quality of migration in shallower parts of the section by summing amplitudes from the flanks of hyperbolic paths that extend beyond the recognizable diffraction amplitudes, thereby introducing noise.

Time-Variant Migration Aperture (TVMA)

The need for a time-variant migration aperture arises from the fundamental kinematic properties of seismic wave propagation in the Earth. As seismic waves travel deeper, the RMS velocity generally increases with time and depth. This increase in velocity causes diffraction hyperbolas to flatten and extend spatially across the seismic section. A fixed aperture, therefore, inherently represents a compromise. It will inevitably be too narrow for effectively imaging deeper, steeply dipping events, leading to their destruction or smearing. Simultaneously, the same fixed aperture will be excessively wide for shallower, less dipping events, introducing unwanted noise by summing irrelevant amplitudes from the hyperbolic flanks.⁵ This dynamic relationship between increasing RMS velocity with time/depth and the corresponding flattening and spatial extension of diffraction hyperbolas is a fundamental kinematic principle that directly dictates the optimal migration aperture. TVMA explicitly leverages this kinematic behavior, ensuring that the summation path aligns optimally with the physical propagation of energy at different depths, leading to superior imaging.

Time-Variant Migration Aperture (TVMA) directly addresses this challenge by allowing the aperture width to dynamically adjust with time (and thus depth). This approach provides a narrower aperture near the surface, where velocities are lower and hyperbolas are more sharply defined, and a wider aperture near the basement, where velocities are higher and hyperbolas are flatter and more extended.

The discussion surrounding aperture size consistently highlights a critical operational consideration: the trade-off between computational cost and image quality. An excessively large aperture increases run time unnecessarily, while an insufficient aperture degrades image quality. TVMA, by dynamically adjusting the aperture to be optimally sized for each time step, aims to achieve a more favorable balance. This approach enables the attainment of superior image quality by utilizing the smallest effective aperture at each depth, potentially leading to more efficient processing for a given quality target compared to a globally large, fixed aperture, or significantly better quality for a similar computational budget as a compromised fixed aperture. This represents a significant step towards optimizing both the output quality and the processing efficiency in complex geological environments.

Methodology and Workflow

Seismic Data Processing

The workflow for this study commenced with rigorous seismic data preconditioning, a crucial initial step essential for enhancing signal quality and effectively attenuating noise prior to the migration process. This preparatory phase involved several critical steps As Below

1. Data Reformat & Resampling
2. Refraction Statics
3. Geometric Spreading Compensation
4. Ground Roll Noise Attenuation (Shot/Receivers' Domain)
5. Preliminary Stacking Velocity Analysis (2000 × 2000 m)
6. Monochromatic Noise Attenuation (Shot Domain)
7. Denoise in CDP Domain

8. Surface Consistent Amplitude Correction (SCAC)
9. Q Compensation (Phase Only)
10. Surface Consistent Deconvolution
11. Velocity Analysis (1000 x1000 m)
12. Residual Static Correction
13. Noise Attenuation (X-spread domain)
14. Radon Demultiple
15. 5D Regularization & Interpolation

Velocity Model Building

An accurate velocity model is of paramount importance for the success of any migration algorithm, directly influencing the precision of reflector positioning and the overall focusing of the final seismic image. The initial velocity model for this study's migration was derived from smoothed stacking velocities, which were obtained from a detailed 500 m velocity analysis grid [Figure 5](#).

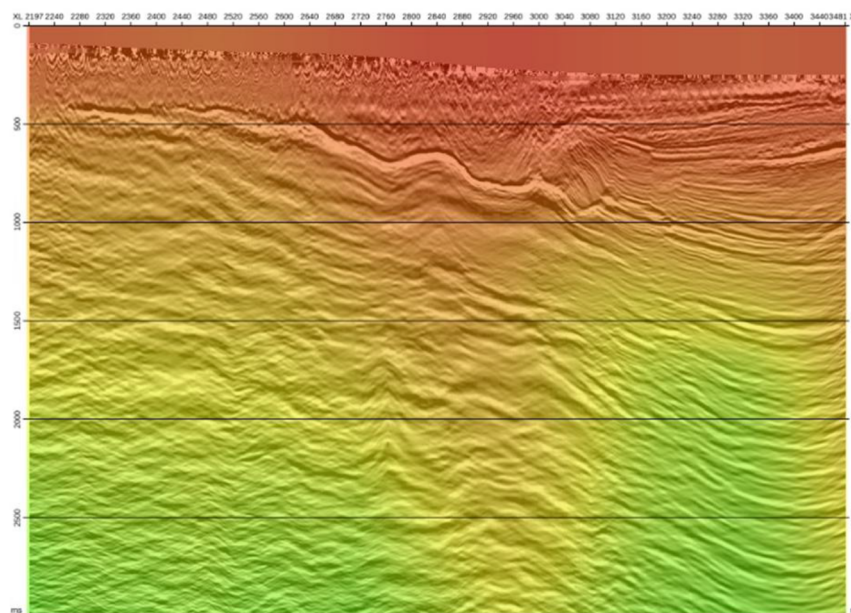


Figure 5—Velocity model for Migration Overlay Seismic Section

This process typically involves primary semblance velocity analysis performed on common midpoint (CMP) gathers to measure Root-Mean-Square (RMS) velocities, followed by gridding and smoothing to create a coherent velocity field.

Dix conversion was applied to these RMS velocities to derive the Interval Velocity (VINT) model.

Optimal Aperture

The determination of optimal aperture functions was a critical phase, achieved through a systematic process of iterative testing and analysis. The primary goal was to maximize the constructive interference of reflection energy while simultaneously minimizing the generation of migration artifacts and background noise.

The success of TVMA in this study, particularly its "specific adaptation and optimization... directly informed by the characteristics of the local dataset and geological environment," highlights a broader trend within the industry.

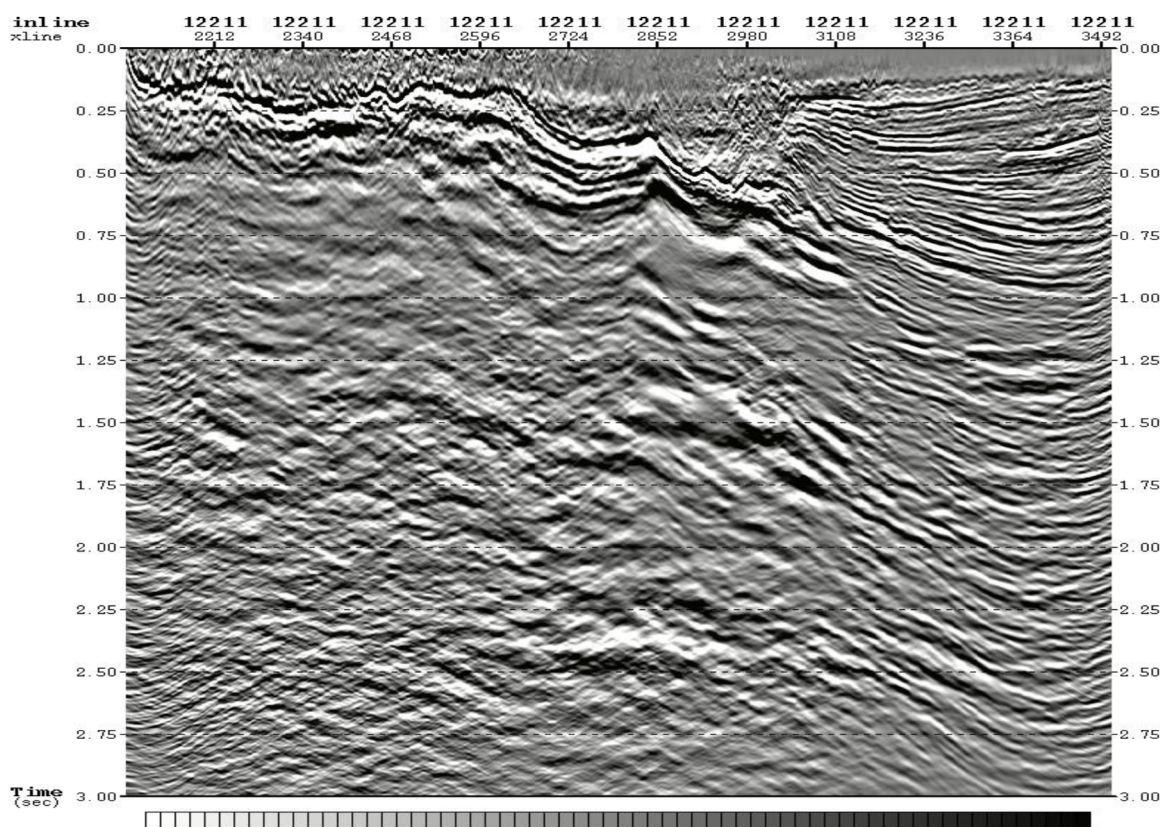


Figure 7—Section Migrated using TVMA Approach

The application of the optimized Time-Variant Migration Aperture (TVMA) Pre-Stack Time Migration (PSTM) technique yielded significant and visually discernible improvements in seismic image quality when compared to results obtained from conventional fixed-aperture PSTM. Visual inspection of the migrated sections clearly revealed enhanced structural definition and improved event continuity across the Area A oil field.

The visual comparison demonstrated improved focusing of reflection energy, particularly noticeable along steeply dipping faults and within complex structural elements. The TVMA results exhibited a significant reduction in migration swing artifacts and background noise, contributing to a cleaner and more interpretable image. Reflector continuity was consistently better, allowing for clearer delineation of subtle structural features that were previously obscured or poorly resolved by fixed-aperture methods.

Conclusions

This study successfully developed and applied a time-variant migration aperture (TVMA) optimization strategy within the Kirchhoff Pre-Stack Time Migration (PSTM) framework. This methodology was specifically tailored for seismic imaging in the geologically complex Area A oil field of the Eastern Desert. The comparative analysis demonstrated that the optimized TVMA PSTM technique yielded substantially improved seismic imaging quality compared to conventional fixed-aperture approaches.

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